

SINPO, Korea, Democratic People's Republic of

WMO: 470460

Lat: 40.033N

Lon: 128.183E

Elev: 19

StdP: 101.10

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.9	-22.1	0.5	-7.9	-19.7	0.7	-7.0	10.2	-3.5	9.1	0.4	1.2	270	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	28.8	24.5	27.7	24.0	26.6	23.3	25.4	28.0	24.7	27.0	24.0	26.1	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.6	27.3	23.9	18.8	26.5	23.2	18.0	25.7	78.3	28.1	75.2	27.1	72.4	26.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	7.8	6.4	-15.2	31.0	1.9	1.1	-16.5	31.8	-17.6	32.4	-18.7	33.0	-20.1	33.8		
			-15.9	26.9	1.8	1.2	-17.1	27.8	-18.2	28.5	-19.2	29.2	-20.4	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-2.7	-1.2	3.5	9.3	14.2	18.6	22.6	23.4	18.9	13.2	5.9	-0.3
	DBStd	9.47	3.16	3.06	3.18	3.18	3.01	2.40	2.71	2.05	2.24	2.88	3.71	3.60
	HDD10.0	1423	395	314	202	51	4	0	0	0	7	131	319	
	HDD18.3	3218	653	548	458	271	131	26	2	19	158	374	578	
	CDD10.0	1611	0	0	1	30	134	258	390	416	268	108	7	0
	CDD18.3	364	0	0	0	0	3	33	133	157	37	1	0	0
	CDH23.3	1646	0	0	0	0	17	73	665	810	80	1	0	0
	CDH26.7	229	0	0	0	0	1	7	114	103	3	0	0	0
Wind	WSAvg	2.1	1.8	1.9	2.4	2.6	2.3	2.0	2.2	1.9	1.8	2.0	2.2	1.9
Precipitation	PrecAvg	719	16	17	19	45	63	77	167	143	90	41	35	19
	PrecMax	1241	60	60	78	207	141	208	369	389	238	83	101	86
	PrecMin	383	5	0	3	6	6	15	12	19	24	1	1	0
	PrecStd	182	11	17	15	39	37	50	87	88	62	25	28	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	8.6	14.7	20.5	25.2	27.8	30.5	30.3	27.3	22.1	17.2	9.7
		MCWB	3.3	4.3	8.9	12.9	16.0	20.5	25.1	25.8	22.1	17.1	11.9	6.0
	2%	DB	4.9	6.5	12.1	18.3	22.7	25.3	29.1	28.8	25.5	20.6	15.1	7.9
		MCWB	1.9	3.4	7.3	11.7	15.3	19.5	24.7	25.0	20.9	16.2	11.0	4.8
	5%	DB	3.7	4.8	10.4	16.5	20.7	23.7	28.0	27.8	24.2	19.4	13.6	6.5
		MCWB	1.0	2.1	6.4	10.7	14.5	19.4	24.3	24.4	19.9	15.4	9.8	3.7
	10%	DB	2.5	3.6	8.6	14.9	19.2	22.5	26.7	26.8	23.1	18.3	12.0	5.1
		MCWB	0.0	1.1	5.2	9.9	14.1	19.1	23.6	23.7	19.4	14.6	8.6	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	5.1	9.7	14.1	18.0	22.2	26.4	26.5	23.6	18.7	13.0	6.9
		MCDB	6.3	7.9	13.6	18.9	21.8	25.3	29.0	29.2	26.4	21.3	15.4	8.7
	2%	WB	2.4	3.7	8.0	12.6	16.6	21.1	25.4	25.6	21.9	17.2	11.7	5.3
		MCDB	4.3	5.9	11.1	16.9	20.9	23.6	28.2	28.1	24.3	19.5	14.7	7.3
	5%	WB	1.4	2.4	6.7	11.3	15.5	20.3	24.6	24.9	20.9	16.2	10.3	4.1
		MCDB	3.3	4.6	9.8	15.5	19.2	22.7	27.3	27.2	23.2	18.7	13.1	6.1
	10%	WB	0.4	1.4	5.6	10.2	14.7	19.5	23.9	24.3	20.2	15.2	9.0	2.8
		MCDB	2.1	3.3	8.3	14.2	18.1	21.8	26.3	26.3	22.4	17.7	11.5	4.9
Mean Daily Temperature Range	5% DB	MDBR	8.0	7.3	6.8	7.6	6.8	5.0	4.4	4.9	6.9	8.1	7.6	7.8
		MCDBR	8.7	8.2	9.1	10.4	10.3	8.2	6.7	6.3	8.1	8.9	9.3	8.5
		MCWBR	7.0	6.1	5.9	5.7	5.1	4.4	3.8	3.7	4.9	6.0	6.7	6.6
	5% WB	MCDBR	8.0	7.5	8.2	9.1	8.4	6.0	6.0	5.3	6.2	7.9	8.5	7.8
		MCWBR	6.7	5.8	5.8	5.7	4.9	3.7	3.8	3.4	4.2	5.8	6.5	6.5
Clear-Sky Solar Irradiance	taub	0.373	0.428	0.501	0.569	0.594	0.634	0.588	0.522	0.465	0.463	0.420	0.372	
	taud	2.088	1.971	1.834	1.736	1.749	1.738	1.910	2.082	2.161	2.023	2.055	2.119	
	Ebn at Noon	749	757	742	722	716	688	716	755	774	712	686	715	
	Edh at Noon	114	148	190	223	225	228	190	156	136	139	118	103	
All-Sky Solar Radiation	RadAvg	2.43	3.34	4.44	5.16	5.48	5.12	4.52	4.49	4.18	3.44	2.40	2.12	
	RadStd	0.16	0.25	0.31	0.40	0.56	0.49	0.58	0.40	0.42	0.29	0.35	0.17	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	+0.43	+0.40	N/A	N/A	+72	N/A	

Nomenclature: See separate page